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Immersal — Visual positioning at scale

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Vision-based 6DoF positioning at scale – turning any camera into a precise positioning sensor.

Immersal Visual Positioning System (VPS), first version released in 2019

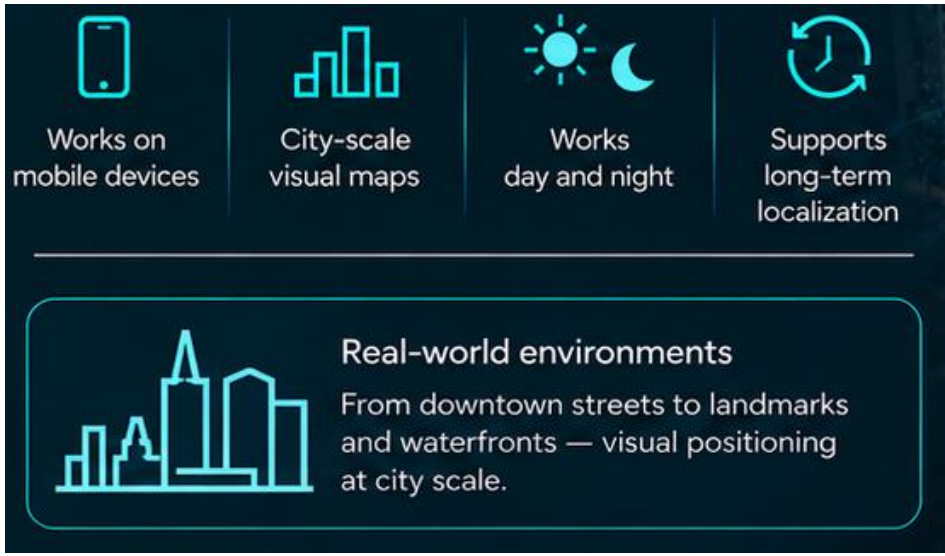
Key features:

- Fast and precise (localisation times below 1s, centimetre-level precision)
- Multi-platform: iOS, Android, macOS, Windows, WebAR, Magic Leap, Ubuntu, HoloLens, XREAL, ROKID, Zappar Mattercraft, 8th Wall, WeChat Mini apps…
- Both on-server (REST API) and on-device localisation (with high-performance native plugins)
- Established product: loved by thousands of developers and trusted by enterprises globally
- Runs on virtually any camera-equipped device – from bodycams and mobile phones to autonomous robots
- Mix and match mapping and positioning devices – combine 360 images with mobile phone images, Matterport or similar scanner, or industrial hardware
- Outperforms Google and Niantic in multiple client-run benchmarks and tech evaluations
- Tailored solutions for special use cases (e.g., landmark localisers)

Next-generation city-scale platform

- State-of-the-art technology: built from the ground up over the last year, our new AI-driven next-gen VPS is already being piloted with enterprise customers
- Cloud-native, GPU-accelerated infrastructure processes maps covering hundreds of square miles with ease
- Uses cost-efficient mapping hardware (e.g., GoPro and Insta360 cameras), reducing the cost of mapping entire cities by an order of magnitude
- Easy to deploy and scale – validated by mapping 50 square miles in Helsinki, Tampere, Zurich, and San Francisco
- Results validated in challenging conditions – across seasons, lighting changes, and times of day
- Our next-gen VPS consistently achieves 99%+ success rates, and delivers precision comparable to RTK GPS
- Competing solutions lack this level of reliability, limiting their viability for industrial applications





Visual positioning in San Francisco



GeoPose support

In city-scale localisation applications, the GeoPose protocol provides a natural way to handle both localisation requests and responses.

- In requests, a GPS prior pose (WGS84 latitude, longitude, ellipsoidal height/altitude) can be used to refine the search area.
- The response to a successful localisation returns the device's 6DoF pose – the position as latitude, longitude, and height, and the rotation as a quaternion in the ENU coordinate system.

Immersal's GeoPose implementation is fully compliant with the OSCP GeoPose protocol version 2

https://api.immersal.com/geoposeoscp

Demo app in Helsinki

